Common native verification

Operator guide | S19b | 2026-10-17 UTC

One fixed command runs 11 existing verification checks. A selected subset stays a subset. No result qualifies whole-service earlier-release recovery, live traffic parity or production activation.

1. Prepare explicit release artifacts

Use a reviewed checkout and the coordinator's Cargo target/build lease. Check disk space and retain the expected build plus a 3 GiB reserve. Build server2 and the six examples below in release mode. Build test executables with the no-run command and use the emitted executable paths.

```
cargo build --locked --release --bin server2 \
  --example loopback_server --example kma_s3_replay \
  --example warnings_smoke --example notice_smoke \
  --example conditional_alert_smoke --example warning_push_smoke

cargo test --locked --release --test summary --test domestic \
  --test state --test runtime_services --no-run --message-format=json
```

Record schema 1, an actual 40-character revision, and artifact role entries containing path, SHA-256 and profile release. Default full execution requires all 11 roles listed by --list. Relative paths resolve from the manifest directory. Files must be executable and match their hashes. Revision/profile are caller declarations, not signed build attestations.

KMA requires the verified examples/kma_s3_replay beside the selected server2 executable. Keep manifests and copied executables in ignored task evidence. The suite does not build or download them.

2. Execute from the repository root

```
python3 server2/tools/native_suite.py --list
python3 server2/tools/native_suite.py \
  --artifacts reports/my-task/artifacts.json --out reports/my-task/run-1
python3 server2/tools/native_suite.py \
  --artifacts reports/my-task/artifacts.json \
  --out reports/my-task/notice-1 --check notice
```

Use a POSIX host with Python 3.9 or later and existing dependencies. The output directory must be new and outside server2/. All selected prerequisites are validated before any child starts. Commands run sequentially with a narrow environment; operator .env, credentials and proxy settings are not inherited.

3. Inspect and retain the report

Inspect selected, full_coverage, each status/layer/limitation, artifact hashes and hashed evidence paths in report.json. PASS applies only to selected checks. full_coverage is true only when all 11 pass. Unselected checks remain NOT_RUN. Preserve original failed outcomes and family reports when retrying.

Coverage and failure handling

Check	Layer and boundary
kma-core	paired-http-replay. Frozen CORE oracle and native S3 HTTP; not an earlier-release rollback.
kma-rich	paired-http-replay. Rich corpus has its own denominator; not an earlier-release rollback.
world	http-coexistence-smoke. Includes a frozen legacy coexistence/rollback fixture; not legacy-free recovery.
warnings	loopback-http-smoke. Synthetic recorded provider and owned S3; no live warning acquisition.
summary-model	library-tests. Model/recorder/lookup tests use in-memory doubles; not HTTP evidence.
summary-http	http-integration-tests. Existing summary_s3 tests only; not all domestic tests or production history.
state	library-and-http-tests. Registration journal/parity/restore/import tests; not whole-service recovery.
notice	loopback-http-smoke. Synthetic owned translation/S3/API peers; no paid translation.
conditional-alert	loopback-http-smoke. Owned weather/OAuth/FCM peers and durable claim; no device send.
warning-push	loopback-http-smoke. Owned warning catalog/geocode/OAuth/FCM peers; no device send.
workers	process-matrix. Current factory: nine phases and three stop triggers; no prior-release qualification.

Exit codes and cleanup

Exit 2 / BLOCKED: invalid prerequisites or output location. Exit 1 / FAIL: executed check, timeout or result validation failed. Logs stay available and later checks still run. Exit zero without expected evidence, missing replay output, partial worker coverage or zero Rust tests is rejected.

A timeout terminates the task-owned process group, including descendants. It does not alter runtime deadlines or retry product failures. Existing output directories are refused rather than overwritten.

CI runs the Python controls and keeps the existing product gates. A controls pass is not an integrated suite execution. Use the full operator command with its explicit artifacts for that evidence.

Selected execution

All 11 checks passed locally. The first run remains FAIL (nine passed; world and conditional alert failed). Bounded verification-fixture repairs retained the original assertions. Production runtime and golden oracles were unchanged.

Executed 2026-10-17T02:49:03.791253+00:00 to 2026-10-17T02:53:37.698843+00:00.

Report SHA-256: `cc43cb227d8d5de443257847bf80d6bf3ce1e5e627140db22632dbf7b48644a6`

```
S19b | actual invocation and retained stdout

python3 server2/tools/native_suite.py
  --artifacts reports/issue-2704-s19b/artifacts-build2.json
  --out reports/issue-2704-s19b/functional-2

PASS kma-core [paired-http-replay]
PASS kma-rich [paired-http-replay]
PASS world [http-coexistence-smoke]
PASS warnings [loopback-http-smoke]
PASS summary-model [library-tests]
PASS summary-http [http-integration-tests]
PASS state [library-and-http-tests]
PASS notice [loopback-http-smoke]
PASS conditional-alert [loopback-http-smoke]
PASS warning-push [loopback-http-smoke]
PASS workers [process-matrix]
PASS native verification: all checks;
  reports/issue-2704-s19b/functional-2/report.json

Report summary (composed from the JSON):
  11/11 PASS | full_coverage=true | process exit=0
  CORE 220 / Rich 92 responses per side | workers 27 cases
  Summary model 27 / HTTP 10 tests | state 18 tests
  No whole-service prior-release or production qualification
Report SHA-256: cc43cb227d8d5de443257847bf80d6bf
               3ce1e5e627140db22632dbf7b48644a6
```

Capture: formatted actual invocation/retained stdout, followed by a labelled JSON summary. It is not a raw terminal attestation. The declared base revision came from a dirty checkout; source and corrected example hashes qualify that declaration in selected evidence.

Editable source: docs/operations/server2-native-verification.md. Selected evidence: docs/evidence/tasks/issue-2704-s19b/verification.md. Workflow: docs/architecture/diagrams/server2-native-verification.html (specification of tool flow).